

## Addendum: Phase 2 Complete Alternate Supplier Master List

### 1. Tier-1 Power Assemblies & Control Boards

- **Aster Power Assemblies & Boreal Power Systems (Tier-1 | M10/M20 Power Assemblies)**
  - **Alternate 1: Flex Ltd.** (<https://flex.com>)
  - **Alternate 2: Delta Electronics** (<https://www.deltaww.com>)
  - *Fitment:* Flex is a massive Tier-1 contract manufacturer capable of building high-voltage power assemblies at scale. Delta is a global leader in power conversion and industrial drive manufacturing, eliminating your reliance on the Aster/Boreal shared-parent choke point.
- **Cobalt Control Electronics (Tier-1 | C10 Control Board)**
  - **Alternate: Plexus Corp** (<https://www.plexus.com>)
  - *Fitment:* Plexus specializes in highly complex PCBA manufacturing specifically for the industrial and energy sectors.
- **Grove Battery Controls & HarborSense Electronics (Tier-1 | B10 Battery Monitor)**
  - **Alternate: Sensata Technologies** (<https://www.sensata.com>)
  - *Fitment:* Sensata is a leading global supplier of mission-critical battery management systems and embedded controllers for industrial and EV infrastructure.

### 2. Tier-1 Passive Components, Thermals & Enclosures

- **Delta Capacitor Works (Tier-1 | C20 DC-Link Capacitor)**
  - **Alternate: TDK (EPCOS)** ([https://www.tdk-electronics.tdk.com/en/film\\_dc\\_link](https://www.tdk-electronics.tdk.com/en/film_dc_link))
  - *Fitment:* TDK Electronics manufactures ultra-compact, high-temperature DC-link film capacitors designed specifically for industrial frequency inverters and solar/EV power supplies.
- **Estuary Thermal Systems (Tier-1 | H20 Cooling Assembly)**
  - **Alternate: Boyd Corporation / Aavid**  
(<https://www.eaton.com/au/en-gb/products/thermal-management-solutions/eaton-and-boyd-thermal.html>)
  - *Fitment:* Eaton's Boyd Thermal division manufactures "Hi-Contact" liquid cold plates and thermal management systems specifically designed for EV batteries and high-power electronics.
- **ForgeLine Enclosures (Tier-1 | H10 Protective Housing)**
  - **Alternate: Hammond Manufacturing** (<https://www.hammfg.com>)
  - *Fitment:* Hammond specializes in heavy-duty, die-cast, and extruded aluminum enclosures engineered to protect sensitive industrial electronics from harsh environments.

### 3. Tier-2 Embedded Electronics & Interconnects

- **Kestrel Microdevices (Tier-2 | Microcontrollers)**
  - **Alternate: NXP Semiconductors** (<https://www.nxp.com/applications/industrial>)
  - *Fitment:* NXP is a foundational supplier of industrial microcontrollers used in energy management, factory automation, and EV supply equipment, providing the exact logic cores needed for the C10 and B10 boards.
- **Rill Connectors (Tier-2 | Connectors)**
  - **Alternate: TE Connectivity** (<https://www.te.com/en/industries/automotive/applications/e-drive-systems.html>)
  - *Fitment:* TE Connectivity delivers high-voltage e-drive interconnection systems designed to handle the heavy current loads of industrial and EV charging systems.

### 4. Tier-2/Tier-3 Advanced Materials (Metals, Resins, Minerals)

- **Lumen Magnetics (Tier-2 | Magnetic Materials)**
  - **Alternate: VACUUMSCHMELZE** (<https://vacuumschmelze.com>)
  - *Fitment:* Global leader in advanced magnetic materials and cores crucial for high-efficiency power conversion.
- **Pine Thermal Metals & Quartz Alloy (Tier-2 | Specialty Alloys)**
  - **Alternate 1 (Thermals): Materion** (<https://materion.com>)
  - **Alternate 2 (Alloys): Carpenter Technology** (<https://www.carpentertechnology.com>)
- **Nacre Foils (Tier-2 | Aluminum/Copper Foils)**
  - **Alternate: Novelis** (<https://www.novelis.com>)
  - *Fitment:* World leader in aluminum rolling and industrial foil packaging.
- **Umber Silicon Carbide (Tier-3 | SiC Wafers) -> Wolfspeed** (<https://www.wolfspeed.com>)
- **Verdant Process Gases (Tier-3 | Gases) -> Linde plc** (<https://www.linde.com>)
- **Warden Copper Foil (Tier-3 | Copper Foil) -> Mitsui Mining & Smelting** (<https://www.mitsui-kinzoku.com>)
- **Xenon Resin (Tier-3 | Resin) -> DuPont** (<https://www.dupont.com>)
- **Yarrow Specialty Minerals (Tier-3 | Minerals) -> Sibelco** (<https://www.sibelco.com>)
- **Zenith Polymer (Tier-3 | Polymers) -> SABIC** (<https://www.sabic.com>)